

predictable: it makes them gray, and only gray.*

As I have said earlier, the world, epistemologically, is literally a different place to a bottom-up empiricist. We don't have the luxury of sitting down to read the equation that governs the universe; we just observe data and make an assumption about what the real process might be, and "calibrate" by adjusting our equation in accordance with additional information. As events present themselves to us, we compare what we see to what we expected to see. It is usually a humbling process, particularly for someone aware of the narrative fallacy, to discover that history runs forward, not backward. As much as one thinks that businessmen have big egos, these people are often humbled by reminders of the differences between decision and results, between precise models and reality.

What I am talking about is opacity, incompleteness of information, the invisibility of the generator of the world. History does not reveal its mind to us—we need to guess what's inside of it.

From Representation to Reality

The above idea links all the parts of this book. While many study psychology, mathematics, or evolutionary theory and look for ways to take it to the bank by applying their ideas to business, I suggest the exact opposite: study the intense, uncharted, humbling uncertainty in the markets as a means to get insights about the nature of randomness that is applicable to psychology, probability, mathematics, decision theory, and even statistical physics. You will see the sneaky manifestations of the narrative fallacy, the ludic fallacy, and the great errors of Platonicity, of going from representation to reality.

When I first met Mandelbrot I asked him why an established scientist like him who should have more valuable things to do with his life would take an interest in such a vulgar topic as finance. I thought that finance and economics were just a place where one learned from various empirical phenomena and filled up one's bank account with *f*** you* cash before leaving for bigger and better things. Mandelbrot's answer was, "*Data*, a gold mine of data." Indeed, everyone forgets that he started in economics before moving on to physics and the geometry of nature. Working with such abundant data humbles us; it provides the intuition of the following error: traveling the road between representation and reality in the wrong direction.

The problem of the *circularity of statistics* (which we can also call the statistical regress argument) is as follows. Say you need past data to discover whether a probability distribution is Gaussian, fractal, or something else. You will need to establish whether you have enough data to back up your claim.